Collision lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test momentum conservation with measured before-and-after velocities.

SUCCESS CRITERIA - I can...

1. Record masses, signs and velocities consistently.
2. Compare total momentum before and after.
3. Discuss external impulse and measurement variation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “momentum” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

momentum: mass times velocity, including direction

external: acting from outside the chosen system

01 Warm-up: bring back an earlier idea

The same object stops from the same speed, but the stopping time is 3 times the original.
Compare average stopping-force magnitudes.

Word help: average = one value representing an interval or a set of values

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total} = m_1v_1 + m_2v_2$; $\Delta p_{total} = p_{total,\,f} - p_{total,\,i}$.
Compare just before/after collision. Use approved carts; record actual trials separately from practice data.

Worked reasoning: Supplied trials give total momenta 6.0 before and 5.8 kg·m/s after. The change is -0.2 kg·m/s. Compare this with measurement uncertainty before attributing it to a new physical effect.

Your turn: Draw a before-and-after momentum comparison and name one possible external interaction.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Supplied case: cart A is 0.50 kg at +0.40 m/s; B is 0.50 kg at rest. After sticking, both move at +0.20 m/s.
Find total momentum before and after.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Cart	Before (m/s)	After (m/s)
A: 0.50 kg	+0.40	+0.20
B: 0.50 kg	0	+0.20

Marzano check - circle: ① ② ③

04 Apply the physics

Two 1 kg lab carts stick together. Before contact their velocities are +2 and 0 m/s; afterward both move at +1 m/s. Complete the table.
Compare total momentum before and after and write the claim the data support.

Sentence starter: Both ... . They differ in ... because ... .

Cart	Mass (kg)	v before (m/s)	v after (m/s)
A	1	2	1
B	1	0	1

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Design a table for three trials with masses and signed velocities before/after.
Name one possible external impulse and one measurement limitation.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A launch has v_x=6 m/s and v_y, i=40 m/s upward.
Use g=10 m/s^2.
Find time to the top and total time back to launch height.
Label separate horizontal and vertical motion.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

07 Exit check

Why must velocities be signed when carts approach each other?

Sentence starter: My claim is … . The diagram/data show …, so … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Group A has momentum change 0.1±0.2; group B has 0.8±0.2 kg·m/s. Which result is consistent with zero under the stated bounds?

Sentence starter: I notice … . This supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③